

The Waiting Game: A Survival Analysis of Fernando Tatis Jr.'s Most Baffling Power Drought

10 Audience Q&As

1. Why survival analysis instead of a simpler model like logistic regression or linear regression?
 - a. Survival analysis is built for time-to-event data and handles right-censored observations well. A player still in an active drought can't be excluded or treated as a non-event without biasing the results. Logistic regression can't model duration, and linear regression doesn't handle censoring.
2. Why did you build the model on one player instead of all power hitters?
 - a. The single-player framing was intentional for this project. It lets the Cox model reflect Tatis' personal survival distribution and makes the anomaly quantification directly interpretable against his own baseline. Expanding to a population model is explicitly called out as the most impactful future extension.
3. Barrel rate has a hazard ratio of 47. Doesn't that seem unrealistically high?
 - a. It's high but interpretable. Barrel rate ranges from 0 to 1 so the coefficient represents the full range from zero barrels to 100% barrel rate. The EDA validates it directly, where barreled balls converted to homeruns at 50% versus 0.5% for non-barreled contact, so the magnitude of the effect is real.
4. Why is pull rate not statistically significant in the Cox model if it showed such a strong descriptive relationship in EDA?
 - a. At the episode level, pull rate and barrel rate are moderately correlated at $r = 0.23$. Pulled balls are more likely to be barreled, so once barrel rate is in the model it absorbs some of the pull rate's explanatory power. Pull rate remains a meaningful descriptive metric even if it doesn't independently reach significance in the model.
5. How did you handle the fact that 2022 is missing from the dataset?
 - a. The 2022 absence is structurally correct since Tatis didn't play that season due to suspension and injury. Episodes are constructed within each season

independently, so there is no cross-season carry-over that would be affected by the off season gap.

6. What does a 3.96% probability at 180 plate appearances actually mean in plain terms?
 - a. It means that if you took 100 stretches of plate appearances where Tatis had his exact 2026 Statcast profile, fewer than 4 of those stretches would reach 180 PAs without a home run. It's unusual but not impossible, and meaningfully less extreme than the unadjusted 0.85% estimate from the Kaplan-Meier curve.
7. What's the difference between the 0.85% from the Kaplan-Meier curve and the 3.96% from the Cox model?
 - a. The Kaplan-Meier estimate treats all episodes equally regardless of covariate values and it's based purely on Tatis' historical drought distribution. The Cox estimate adjusts for the fact that his 2026 Statcast profile is genuinely worse for home run production. The gap between the two numbers quantifies how much of the drought his changed mechanics explain.
8. Could this be an injury situation that your model isn't capturing?
 - a. Yes, that is possible and it's an honest limitation. The model only uses Statcast batted ball and plate discipline metrics so it has no visibility into physical health. However, the stable exit velocity is somewhat reassuring since injured hitters typically show reduced bat speed and contact quality. The launch angle collapse is more consistent with a mechanical or stance-related change than with injury.
9. You mentioned the proportional hazards assumption but what would you have done if it was violated?
 - a. If Schoenfeld residuals had flagged a violation, the options would have been stratification (splitting the model by the offending covariate), introducing a time-varying coefficient for that covariate, or switching to an accelerated failure time model that doesn't require the PH assumption. Fortunately none of that was necessary since all six covariates passed.
10. When do you predict Tatis will hit his first home run?
 - a. The model doesn't produce a specific date prediction, but a survival probability at any given plate appearance count. What it can say is that under his 2026 profile, the median expected drought is 26 plate appearances, meaning he's already well past the point where 50% of comparable stretches would have ended. Every plate appearance from here is increasingly unlikely territory.